THE SPECTRE KNOT: ONE ENDLESS STRAND ON AN APERIODIC MONOTILE

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Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane, but only non-periodically. We decorate its straight-edged form Tile(1, 1) with fourteen ribbon pieces, the same on every tile, that continue straight across every tile edge. We prove that in every tiling by copies of Tile(1, 1) of one handedness whose supertile hierarchy covers the plane no strand of ribbon is closed, and that in the tiling grown from a nested chain of Δ-supertiles all the pieces join into a single strand, a two-way infinite path through every tile; we do not claim this for other tilings. Every supertile connects the ribbon ends on its boundary in the same pattern; ribbons zigzag along a shared boundary, so lengthening it by two edges changes nothing; and the side lengths at every level are reached from those at level 1 in such steps. So finitely many computations at level 1 settle every level. The combinatorial core is formalised in the Lean proof assistant. The author used an AI system to produce the proof, the programs, the formalisation and this text. This paper has not been peer reviewed (Section 1).

1. Introduction

A *Celtic knot* is a pattern of interlaced ribbons [2, 6]; the number of strands in finite designs of this kind has been studied, for instance, by Fisher and Mellor [9]. A *knot tile* is a polygon carrying ribbon segments that end on its edges, so that copies of the tile, placed edge to edge, link up into longer ribbons; knot mosaics [23] assemble knot diagrams in this way from a set of square tiles. We ask for a knot tile whose ribbons, in a tiling of the whole plane, form one endless strand. In a periodic tiling this cannot happen: if the pieces formed one strand, a period τ would shift it along itself by some $k \neq 0$ (τ^2 moves every tile, so τ cannot reverse it), leaving $|k|$ orbits of pieces under $\langle\tau\rangle$ but infinitely many of tiles, as $\mathbb{R}^2/\langle\tau\rangle$ has infinite area. So we look at a tile that tiles only non-periodically.

Socolar and Taylor [30] had found a hexagonal tile that, with its mirror image, tiles only non-periodically when its markings are matched. In 2023 Smith, Myers, Kaplan and Goodman-Strauss described the *hat*, an aperiodic monotile [28], and then the *Spectre* [29]: a shape that tiles the plane, admits only non-periodic tilings, and admits only tilings in which all tiles have the same handedness, even when reflections are allowed [29, Thm. 2.2]. The Spectre is a curved-edge form of the polygon Tile(1, 1) (Fig. 1, Section 2). For background on tilings and substitution see [16, 1].

We give a knot tile on the Spectre that is *mono-stranded*: in one explicit tiling all its pieces form a single strand (Theorem B), and no tiling whose hierarchy covers the plane contains a closed strand (Theorem A). The decoration (Fig. 1) puts one crossing point on each of the fourteen edges and joins the 28 ribbon ends inside the tile by fourteen pieces: twelve *hooks*, each turning back round one vertex, and two pieces that cross each other inside the tile. Across an edge, the ribbons continue straight through the crossing point.

Every tiling by Tile(1, 1) with tiles of one handedness (a *homochiral* tiling) groups uniquely into a hierarchy of larger and larger *supertiles* [29, Thm. 2.2]. We say that the hierarchy *covers the plane* if the supertiles containing some tile exhaust the plane. This does not depend on the tile: once a supertile containing one tile contains another, the two tiles have the same

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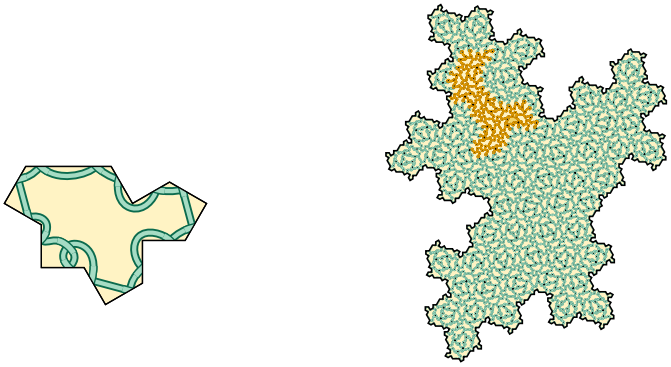


FIGURE 1. Left: the tile Tile(1, 1) with its fourteen ribbon pieces. Right: a supertile of 559 tiles (the block N_3 of Section 8). The original tile N_0 is shaded; all fourteen of its pieces already lie on a single arc of this block, drawn in orange.

supertiles from that level on. Otherwise the tiling splits into two or more *infinite supertiles*, unions of nested chains of supertiles, separated by a *fault* (Section 10).

Theorem A (The Daedalus Theorem)**.** *In every homochiral tiling of the plane by* Tile(1, 1) *whose supertile hierarchy covers the plane, with every tile carrying the decoration of Fig. 1, there is no closed strand: every strand is a two-way infinite path of ribbon pieces.*

Our second result is proved for one tiling, the *nest tiling* $\mathcal{N}$, grown from a single tile N_0 by repeatedly taking the Δ-supertile N_{b+1} that has N_b as its child number 1 (Section 8). In the language of substitution tilings, $\mathcal{N}$ is generated from one tile by iterating the substitution about it, a fixed point of the substitution (in the combinatorial sense; compare [1]); in the tile addresses of [31, §2.8] (after [14]), N_0 has the address $\Delta \overset{1}{\leftarrow} \Delta \overset{1}{\leftarrow} \Delta \overset{1}{\leftarrow} \cdots$.

Theorem B (The Ariadne Theorem)**.** *In the nest tiling $\mathcal{N}$, with every tile carrying the decoration of Fig. 1, all the ribbon pieces of all the tiles form a single strand. It is a two-way infinite path that passes through every tile. More precisely, for every $b \geq 1$, all the ribbon arcs of the block N_b lie on a single arc of N_{b+4}, and of N_{b+3} when b is even.*

We make no claim about the number of strands in any other tiling: computations indicate that in at least one other tiling whose hierarchy covers the plane the pieces form three strands; and tilings with faults are covered by neither theorem (Section 10).

The method. The proof has three ideas.

(1) *Black boxes.* Seen from outside, a supertile is just a list of which ribbon end on its boundary is joined to which. Every supertile of every kind and level $m \geq 1$ has the same *signature*: a hook at every boundary vertex but two adjacent ones, A and B, where the two ribbons cross over; and no closed strand inside (Definition 4.3, Theorem 7.6).
(2) *Zigzags.* When two such boxes are glued along a common boundary path, the ribbons zigzag along it, and the outcome depends only on the parity of its length (Lemma 5.1). So lengthening a shared path by two edges changes nothing (Lemma 7.1).
(3) *Fixed parities.* The supertiles are combinatorially hexagons and compose by the substitution of [29, Fig. 5.1]. Their side lengths are given by Fibonacci numbers (Proposition 6.2); they only grow with the level, and their parities never change. So the side lengths at any level are reached from those at level 1 by lengthening, two edges at a time (Lemma 7.2), and a finite computation at level 1 gives the signature at every level.

The single strand (Theorem 9.1) follows the same pattern. Where the ribbons of a block go one level up depends only on where they leave it, for most of its rim only on the side

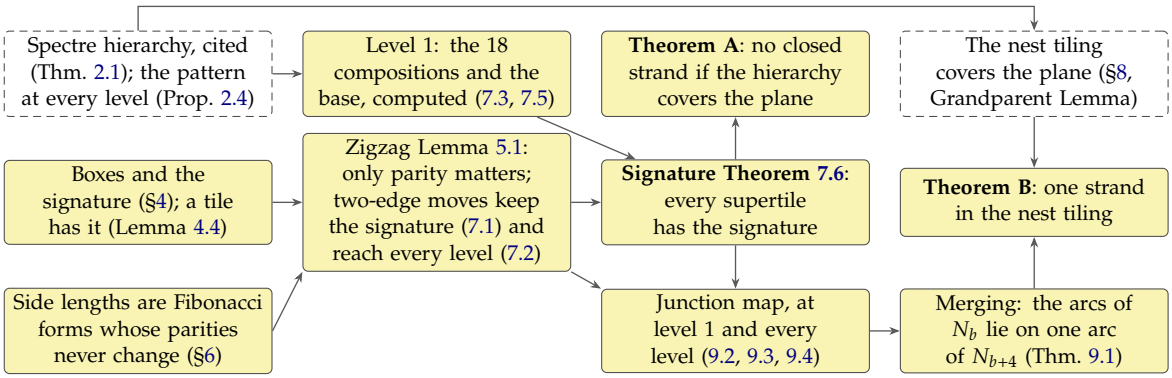


FIGURE 2. The route of the proof. Shaded: checked by the Lean formalisation in a combinatorial form, including the computations at level 1 (Appendix B). Dashed: the link with actual tilings, cited from [29] or argued in words.

(Lemma 9.3); this map is computed at level 1 and carried to every level by the same lengthening. Followed three or four levels up, it puts every arc of a block on one strand. The finite computations (the level-1 data, the agreement with [29, Figs. 4.1, 5.1], and Computations 7.3 and 9.2 on the 18 compositions of level-1 supertiles) are exact; the last two are repeated in the kernel of the Lean proof assistant (Appendix A). Figure 2 shows how the steps fit together, and which of them Lean checks.

Related work. Curves drawn on tiles and joined across edges, each closed or two-way infinite as in Lemma 3.4, have been studied for random Truchet tilings [25, 17] and a deterministic family of them, where a renormalisation, akin to our induction on supertiles, shows that for almost every parameter almost every curve is closed [18]; for the mirror curves of Celtic, Tchokwe and Tamil designs [13, 19], where a design drawn as a single line is called *monolinear*, and a rectangular grid of $m \times n$ cells with no inner mirrors carries $\gcd(m, n)$ curves [13]; for the Conway–Penrose arcs on Penrose tiles, where at most two curves of each colour fail to close [12, Ch. 1] (see also [10]); and for girih interlace on Penrose tilings, whose strands close into loops and knots [4, 5, 11]. Single curves through finite patches have been built as Hamiltonian cycles, through every vertex of arbitrarily large regions of Ammann–Beenker tilings [26] and through every tile of finite Penrose and Ammann–Beenker patches [22], and as space-filling curves from aperiodic substitution tilings [24]. Closest to ours, an online project [3] decorates selected edges of Tile(1, 1) and describes, in one nested tiling, a single non-crossing curve through every tile, with a machine check of its combinatorial core. Here every edge carries a ribbon end and the pieces cross; in our drawing they cross only transversally and in pairs (Appendix A), so over and under can alternate by the checkerboard rule [7], and the single strand becomes an alternating diagram of a line, as in Celtic knotwork, whose designs are alternating and give every alternating link projection up to topological equivalence [15]. We are not aware of earlier full knotwork decorations that form a single strand.

Use of AI and verification status. This work is an experiment in using AI for mathematical exploration: the author used an AI system (Claude, by Anthropic) at every stage; it did the detailed work, including the programs, the proofs, the Lean formalisation and this text, and the author directed, guided and corrected it. The author is responsible for the paper: for the correctness of its arguments and results, and for the completeness and accuracy of its citations of prior work. The AI system is not an author. Lean checks the combinatorial core of the proof; the link with actual tilings is argued in words. Most of the proof has not been reviewed by a human, and the paper has not been peer reviewed. Appendix B gives the full account: how the work was done, what the formalisation checks, and the tools used.

parent T	0	1	2	3	4	5	6	7
Γ	Π	Δ	Θ	Σ	Ξ	Φ	Γ	
Δ	Ξ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Θ	Ψ	Δ	Π	Φ	Σ	Π	Φ	Γ
Λ	Ψ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Ξ	Ψ	Δ	Π	Φ	Σ	Ψ	Φ	Γ
Π	Ψ	Δ	Ξ	Φ	Σ	Ψ	Φ	Γ
Σ	Ξ	Δ	Ξ	Φ	Σ	Π	Λ	Γ
Φ	Ψ	Δ	Ψ	Φ	Σ	Π	Φ	Γ
Ψ	Ψ	Δ	Ψ	Φ	Σ	Ψ	Φ	Γ
position in [29, Fig. 5.1]	L	D	B	R	S	TR	TL	G

TABLE 1. The children of each kind of supertile. For $T = \Gamma$ there is no child at position B, and the positions of children $2, \dots, 6$ are R, S, TR, TL, G. The positions name the places in the eight-hexagon layout of [29, Fig. 5.1]: L left, D the Δ, B bottom, R right, S the Σ, TR top right, TL top left, G the Γ.

2. SPECTRE TILINGS

2.1. **The tile.** Let $\rho = e^{i\pi/6}$. The polygon Tile$(1,1)$ has fourteen unit edges; we label its vertices $v_0, \dots, v_{13}$ counter-clockwise and its edges $e_i = v_i v_{i+1}$ (indices mod 14), with $v_0 = 0$ and

$$v_{i+1} - v_i = \rho^{k_i}, \qquad (k_0, \dots, k_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

All vertices lie in $\mathbb{Z}[\rho]$. This is the equilateral member of the family Tile(a, b) [29, §§1–2], with its long side split into the collinear edges e_9, e_{10}; it is the mirror image of the polygon drawn in [29, Fig. 1.1], and has the handedness of the single Spectre and of the Mystic in [29, Fig. 2.1]. *Spectres* are tiles, obtained for instance by replacing every edge of Tile$(1,1)$ by a suitable curve, whose tilings correspond exactly, with the same isometries, to the *homochiral* tilings by Tile$(1,1)$, in which any two tiles are related by a rotation and a translation [29, §2, Lemma 2.1]; so what [29] proves for tilings by Spectres holds for homochiral tilings by Tile$(1,1)$ [29, p. 5]. Such tilings exist [29, Thm. 2.2], and in each, tiles meet full edge to full edge [29, proof of Lemma 2.1]. Reflecting the plane if necessary (with the decoration below, which maps strands to strands), we assume that every tile is the image of Tile$(1,1)$ under a rotation and a translation. The tile has no non-trivial symmetry, proper or improper (its cyclic sequence of edge directions has none), so an isometry between two tiles is unique.

2.2. **Supertiles.** We use the hierarchy of [29], with nine *kinds* $\mathcal{K} = \{\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi\}$. A *level*-0 *supertile* of kind $K \neq \Gamma$ is a single tile; one of kind Γ is a *Mystic* [29, Fig. 2.1], the pair of tiles $\{\text{Tile}(1,1),\ g(\text{Tile}(1,1))\}$ with $g(z) = \rho z + v_8$, or a congruent copy, with *first* tile Tile$(1,1)$. The first tile is determined by the pair, since the Mystic has no non-trivial proper symmetry: one exchanging its tiles would be an involution taking Tile$(1,1)$ to $g(\text{Tile}(1,1))$, but the only proper motion doing so is g, a rotation by $30°$. A level-$(m+1)$ supertile is the union of level-m supertiles, its *children*: eight, or seven for kind Γ, of the kinds listed in Table 1. A level-1 supertile has the kind of its cluster in [29, Fig. 4.1], which is determined by the positions of its six vertices [29, proof of Thm. 4.1]; a level-$(m+1)$ supertile, $m \geq 1$, has kind T if its children form the panel T of [29, Fig. 5.1]. Children are numbered as in Fig. 3 at level 1, and by their positions in the last row of Table 1 above. A single tile gets the kind of its position in its level-1 supertile; only the distinction between Γ and the other kinds is used at level 0. Child 1 of every supertile is a Δ and the last child a Γ; at levels $m \geq 1$ a Δ occurs only as child 1.

The level-1 supertiles are the Spectre clusters (9 tiles) and the Mystic cluster (8 tiles, kind Γ) of [29, Fig. 2.1], with the markings of [29, Fig. 4.1], up to reflection ([29, Fig. 2.1] draws them with tiles of the opposite handedness to Tile$(1,1)$, Fig. 3 with that of Tile$(1,1)$). They are

combinatorially the marked hexagons of [29, Fig. 4.2], and the level-n supertiles of hexagons of [29, §5] are our level-$(n+1)$ supertiles.

The following theorem restates, in our language, results of [29].

Theorem 2.1 (Smith, Myers, Kaplan, Goodman-Strauss [29])**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$.

(a) [29, Thms. 2.2, 3.1, 4.1, §5, Fig. 2.1] *The tiles of $\mathcal{T}$ group uniquely into level-1 supertiles. Each contains one Mystic, in the position shown in* [29, Fig. 2.1]*; this Mystic and the other tiles, taken singly, are the level-0 supertiles inside it. For every $m \geq 1$ the level-m supertiles of $\mathcal{T}$ group uniquely into level-$(m+1)$ supertiles. Each tile lies in a unique nested sequence $X_0 \subset X_1 \subset \cdots$ with X_m a level-m supertile.*

(b) [29, Thms. 3.1, 4.1, 5.1, §5] *For every $m \geq 1$, the tiling of the plane by the level-m supertiles of $\mathcal{T}$ is combinatorially equivalent (homeomorphic as a topological complex* [29, §2.2]*) to a tiling by regular hexagons, each level-m supertile of kind K corresponding to a hexagon marked as the hexagon K of* [29, Fig. 4.2] *for even m, and as its mirror image for odd m, adjacent supertiles corresponding to hexagons sharing an edge with matching labels; and the grouping into level-$(m+1)$ supertiles corresponds, up to reflection, to the grouping of hexagons shown in* [29, Fig. 5.1].

In (a), [29, Thm. 4.1] gives the clusters and [29, Thm. 2.2] the uniqueness at every level, which we use only to speak of *the* hierarchy of $\mathcal{T}$. In (b), the handedness is that of our level-1 supertiles, the mirror images of the clusters in [29, Figs. 2.1, 4.1] (see the base case of $(*)$ in the proof of Proposition 2.4); it alternates with m by [29, Thm. 5.1].

In particular every level-m supertile ($m \geq 1$) is a closed topological disc whose boundary, its *rim*, is a simple closed polygon made of tile edges. Its *vertices* are the six points of the rim where it meets two other level-m supertiles, and its *sides* are the six arcs of the rim between consecutive vertices. (The tile vertices on a rim are called *rim vertices*; for a supertile of level ≥ 1, "vertex" alone always means one of the six.) A vertex of one supertile that lies on the rim of another is a vertex of that one too, since exactly three supertiles meet there. Each side is shared, as a path of tile edges, with exactly one neighbouring supertile.

2.3. **Own orientation, side numbering and the crossover corner.** By [29, Thm. 5.1], the supertiles composed from a tiling by marked hexagons are combinatorially equivalent to the *mirror images* of those hexagons. So the cyclic order of the labelled sides reverses at each level, and we orient supertiles alternately.

Definition 2.2 (Own orientation)**.** The *own orientation* of a level-m supertile is counterclockwise if m is even and clockwise if m is odd. We write $\omega_m = (-1)^m$.

Thus a child and its parent are always oriented oppositely. Each supertile of level $m \geq 1$ has a distinguished rim vertex A, its *crossover corner*, defined recursively: for a level-1 supertile, A is the vertex v_0 of the first tile of its Mystic; for a level-$(m+1)$ supertile, A is the crossover corner of its Γ child. (That this point lies on the rim is computed at level 1 and follows from Proposition 2.4(d) at higher levels.) The second corner B, adjacent to A, appears in Theorem 7.6; we call A and B the *crossover corners*. The vertices of a level-m supertile ($m \geq 1$) are numbered $0, \ldots, 5$ in own order, starting with the first vertex at or after A; side j runs from vertex j to vertex $j+1$ (Fig. 3).

2.4. **The composition pattern.** By Theorem 2.1(b), every level of the hierarchy is assembled in the same combinatorial way. We record this in our numbering.

Definition 2.3 (Composition pattern)**.** The *composition pattern* gives, for each parent kind T, the kinds $K_0, \ldots, K_{n_T-1}$ of the children (Table 1); for each side of each child, the sibling's side across it, or ∂ if it lies on the parent's rim (Table 2); and for each side of the parent, the children's sides that make it up, in the parent's own order (Table 3). Figure 4 draws it.

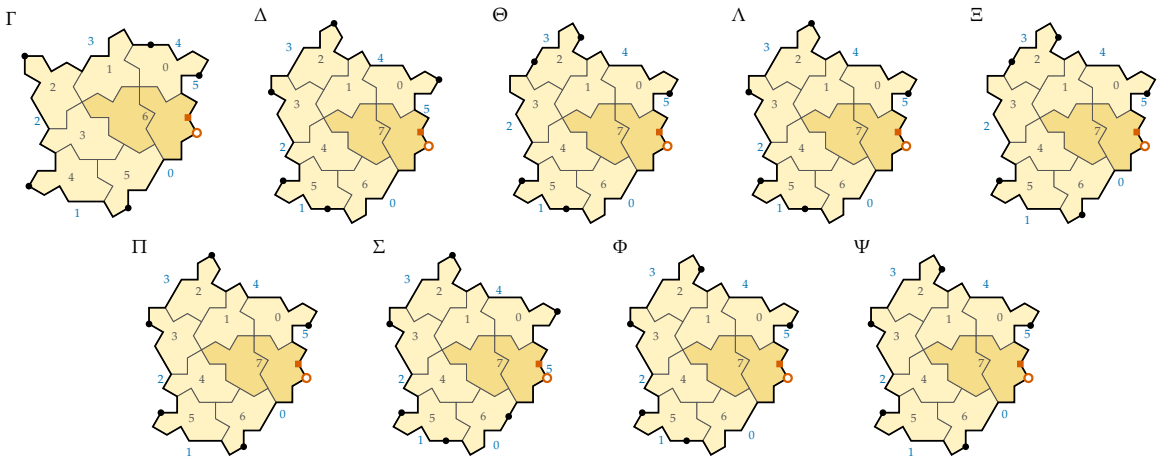


FIGURE 3. The nine level-1 supertiles, with true handedness. The Mystic is the darker pair of tiles. Grey numbers: the children (Table 1); black dots: the six vertices; blue numbers: the sides, numbered clockwise (the own orientation at level 1); open circle: the crossover corner A, which is vertex 0 except for Σ; filled square: the corner B (Theorem 7.6). For Σ, A lies inside side 5.

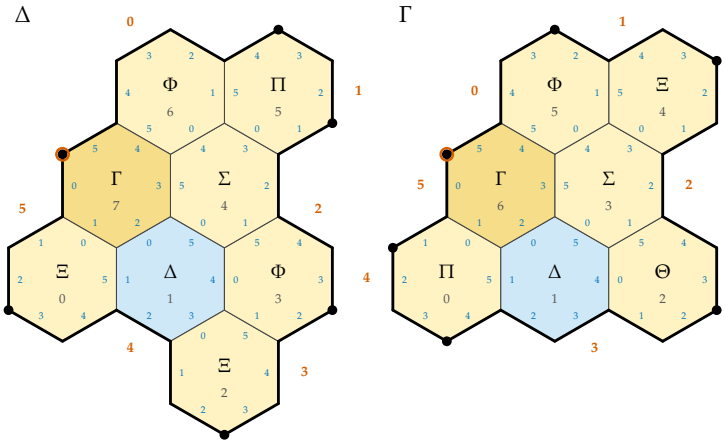


FIGURE 4. The composition pattern for Δ (left; the same for every kind but Γ, with the children's kinds of Table 1) and for Γ (right), in the layout of [29, Fig. 5.1]. Each hexagon is a child: kind, number (grey), sides 0–5 (blue). Thick black: the parent's rim; dots: its vertices; vermilion: its sides; circled: its crossover corner, vertex 0 of the Γ child. Children's sides are numbered in one rotational sense, the parent's in the other.

Proposition 2.4 (The pattern holds at every level)**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$*, let $m \geq 1$, and let Y be a level-$(m+1)$ supertile of $\mathcal{T}$ of kind T, with children $X_0, \ldots, X_{n_T-1}$. Then:*

(a) *X_c has kind K_c (Table 1);*
(b) *side j of X_c is the same path of tile edges as side j' of $X_{c'}$, traversed in the opposite direction, whenever Table 2 lists $c'.j'$, and it lies on the rim of Y when the table lists ∂;*
(c) *side J of Y is the concatenation of the children's sides listed in Table 3, and vertex J of Y is the vertex $j+1$ of the child c whose side $c.j$ is listed first;*
(d) *the crossover corner A of Y is vertex 0 of its Γ child, which is vertex 0 of Y if $T \neq \Sigma$, and lies inside side 5 of Y if $T = \Sigma$.*

every parent kind except Γ						
child	side 0	1	2	3	4	5
0	7.1	∂	∂	∂	∂	1.1
1	7.2	0.5	∂	2.0	3.0	4.0
2	1.3	∂	∂	∂	∂	3.1
3	1.4	2.5	∂	∂	∂	4.1
4	1.5	3.5	∂	5.0	6.0	7.3
5	4.3	∂	∂	∂	∂	6.1
6	4.4	5.5	∂	∂	∂	7.4
7	∂	0.0	1.0	4.5	6.5	∂

parent kind Γ						
child	side 0	1	2	3	4	5
0	6.1	∂	∂	∂	∂	1.1
1	6.2	0.5	∂	∂	2.0	3.0
2	1.4	∂	∂	∂	∂	3.1
3	1.5	2.5	∂	4.0	5.0	6.3
4	3.3	∂	∂	∂	∂	5.1
5	3.4	4.5	∂	∂	∂	6.4
6	∂	0.0	1.0	3.5	5.5	∂

TABLE 2. Adjacency of the children's sides: the entry $c'.j'$ in row c, column j means that side j of child c is side j' of child c'; ∂ means that it lies on the parent's rim.

Eight children (every $T \neq \Gamma$): the parent's rim, in own order from vertex 0 of child 7.

position	0	1	2	3	4	5	6	7	8	9	10
piece	7.5	6.4	6.3	6.2	5.4	5.3	5.2	5.1	4.2	3.4	3.3
position	11	12	13	14	15	16	17	18	19	20	21
piece	3.2	2.4	2.3	2.2	2.1	1.2	0.4	0.3	0.2	0.1	7.0

parent	first position of side					
	0	1	2	3	4	5
Δ	0	5	7	11	14	19
Θ	0	5	7	13	15	20
Λ	0	5	7	11	14	20
Ξ	0	3	7	13	15	20
Π	0	3	7	11	14	20
Σ	2	5	7	11	14	19
Φ	0	5	7	11	15	20
Ψ	0	3	7	11	15	20

Seven children ($T = \Gamma$), from vertex 0 of child 6; sides $0, \ldots, 5$ begin at positions $0, 3, 6, 11, 16, 18$.

position	0	1	2	3	4	5	6	7	8	9
piece	6.5	5.4	5.3	5.2	4.4	4.3	4.2	4.1	3.2	2.4
position	10	11	12	13	14	15	16	17	18	19
piece	2.3	2.2	2.1	1.3	1.2	0.4	0.3	0.2	0.1	6.0

TABLE 3. The parent's sides as concatenations of children's sides. A piece $c.j$ is side j of child c, which the parent's own orientation traverses from vertex $j+1$ to vertex j of the child. Side J of the parent consists of the pieces from its first position up to the first position of side $J+1$ (cyclically). For $T \neq \Sigma$ the parent's side 0 begins at vertex 0 of the Γ child, which is the parent's crossover corner. For $T = \Sigma$ the crossover corner lies inside side 5 (positions 19–1), after the pieces 0.2, 0.1, 7.0.

Justification. By Theorem 2.1(b), the level-m supertiles inside Y are combinatorially the hexagons of the panel T of [29, Fig. 5.1], with hexagon edges as sides and hexagon vertices as vertices. We use the dictionary

(∗) side j of a level-m supertile of kind K carries the label of the edge of hexagon K in direction W, SW, SE, E, NE, NW for $j = 0, \ldots, 5$ (Table 4); its crossover corner A is vertex 0 if $K \neq \Sigma$; if $K = \Sigma$, A is not a vertex, and lies in the relative interior of side 5,

and show (∗) by induction on m. For $m = 1$ it is a computation on the level-1 supertiles, whose vertices and corner A are marked in Fig. 3. Drawn with the edges in odd directions lengthened to $\sqrt{3}$, which turns the tiles into hats and turtles [29, §3], and in the handedness of [29, Fig. 4.1], they coincide with the nine marked clusters there: the same outlines, the vertices at the marked dots, the labels of Table 4 where that figure has them, and A the dot

between the sides labelled W and NW (for Σ, a point of the NW side), on the light tile of the Mystic (checked by a script; Appendix A). Suppose $(*)$ holds for the children of Y. The panel does not label its inner hexagons, but the matching rules fix the rotation of each, and there is exactly one consistent choice (Appendix A). With it, Tables 2 and 3 transcribe the panel, including its six marked vertices; this gives (a)–(c). The crossover corner of Y is that of its Γ child, the child's vertex 0 by $(*)$, which Table 3 places at the start of the parent's side 0 if $T \neq \Sigma$ and inside side 5 if $T = \Sigma$; this is (d). Finally, the labels that the panel assigns to the parent's sides, numbered from this point in the parent's own order, are those of Table 4 for every T. So $(*)$ holds for Y, and in particular A lies on the rim. □

Remark 2.5 (Computer cross-checks of the pattern). Since Tables 2 and 3 carry the whole weight of the citation, we also checked them on actual supertiles (evidence, not used in the proof): the patterns of the compositions $m \longrightarrow m+1$, $m \leq 7$, extracted from them equal the tables, and Kaplan's implementation of the substitution [27] gives Table 1 and the same supertiles, tile for tile, up to level 4.

Lemma 2.6 (Rigidity)**.** *Any two level-m supertiles of the same kind, in possibly different homochiral tilings, are related by a rotation and translation that maps descendants to descendants with their numbers (children, their children, and so on, down to the tiles), and it is the only such motion. For $m \geq 1$ it maps the six vertices, the sides and the corner A of one onto those of the other.*

Proof. Induction on m. For $m = 0$ this is clear, since a tile and a Mystic have no non-trivial proper symmetry. For $m = 1$, the level-1 supertiles are congruent copies of the clusters of Fig. 3 [29, Fig. 2.1, §4], with their vertices at the points marked in [29, Fig. 4.1], their children numbered as in Fig. 3, and A fixed by the Mystic. Let $m \geq 1$ and let Y, Y' be level-$(m+1)$ supertiles of kind T. By induction, child c of Y is mapped onto child c of Y', with its numbered vertices, by a unique proper motion g_c. If side j of child c is glued to side j' of child c', then by Proposition 2.4(b), in Y and in Y', vertices j and $j+1$ of c are vertices $j'+1$ and j' of c'; so g_c and $g_{c'}$ agree on two distinct points and are equal. The children's adjacency graph is connected, so all the g_c are one motion g; it maps descendants to descendants with their numbers and, by Proposition 2.4(c),(d), the vertices, sides and corner A of Y onto those of Y'. A proper motion that maps descendants to descendants maps the first tile of the Mystic at the bottom of the chain of last (Γ) children to its counterpart, so it equals g, since a tile has no symmetry. □

Write $E_m(K)$ for the number of rim edges (equally, rim vertices) of a level-m supertile of kind K, and $\ell_m(K, j)$ for the number of edges of its side j. By Lemma 2.6 these numbers are well defined.

3. The decoration and its strands

3.1. **Ports and pieces.** Let t be a tile and e one of its edges, with endpoints x and y. The tile has two *ports* on e, the port *near* x and the port *near* y; formally, they are the pairs (e, x) and (e, y). In the drawing (Fig. 5), e carries one crossing point, and the two ribbon ends of t that meet it, one running off towards x and one towards y, are these two ports. For a vertex v_i of the tile, write

$$\mathrm{R}(v_i) = \text{the port on } e_i \text{ near } v_i, \qquad \mathrm{L}(v_i) = \text{the port on } e_{i-1} \text{ near } v_i,$$

the ports just after and just before the corner v_i in counter-clockwise order. Each of the 28 ports is $\mathrm{R}(v_i)$ or $\mathrm{L}(v_i)$ for exactly one i.

Definition 3.1 (The decoration)**.** The *decoration* joins the 28 ports of Tile$(1,1)$ in pairs by fourteen *pieces*:

$$\underbrace{\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i) \quad (2 \leq i \leq 13)}_{\text{twelve hooks}}, \qquad \underbrace{\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1), \qquad \mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)}_{\text{the crossover}}.$$

Every tile of a homochiral tiling carries the image of this decoration.

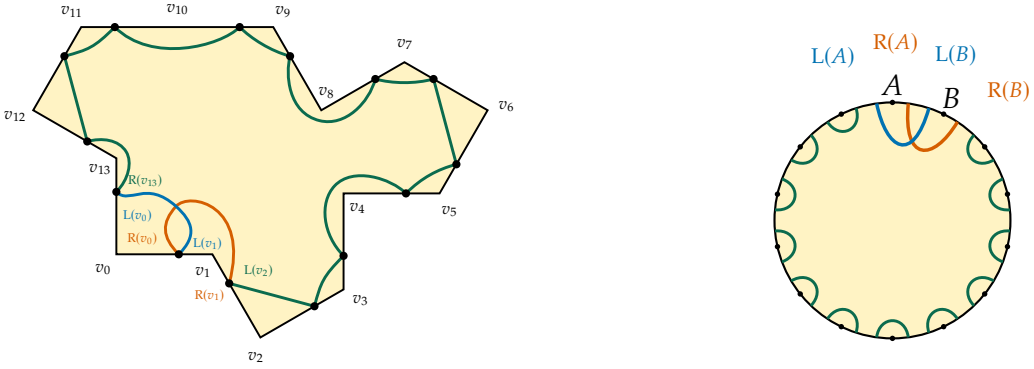


FIGURE 5. Left: the decoration (centre lines of the ribbons of Fig. 1). Each edge carries one crossing point (dot); on e_i the two pieces ending there leave from the ports $\mathrm{R}(v_i)$ (towards v_i) and $\mathrm{L}(v_{i+1})$ (towards v_{i+1}). Green: the twelve hooks $\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i)$; vermilion $\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1)$ and blue $\mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)$: the crossover at $A = v_0$, $B = v_1$. Right: the *signature* at (A, B) (Definition 4.3) on a rim of 14 vertices oriented clockwise: a hook at every rim vertex but A and B, and the crossover arcs $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$, $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$. The tile has it (Lemma 4.4); so does every supertile of level $m \geq 1$, seen as a black box (Theorem 7.6).

Each hook turns back round one vertex; the two crossover pieces cross once inside the tile. Only the pairing of Definition 3.1 matters for the theorems; any drawing that realises it may be used.

3.2. **The straight-through rule.** Let $\mathcal{T}$ be a homochiral tiling, and let e be an edge shared by tiles t and t', with endpoints x and y.

Definition 3.2 (Straight-through rule)**.** The port of t on e near x is *joined* to the port of t' on e near y, and the port of t near y to the port of t' near x.

This is what a ribbon does when it continues straight through the crossing point. In our drawing the crossing point of e_i is at distance $0.5+0.15(-1)^i$ from v_i, so the two tiles' crossing points on e coincide: e_i is always glued to some e_j with $i + j$ odd, by the vertex colouring of [29, proof of Lemma 2.1]. The proofs use only Definitions 3.1 and 3.2.

3.3. **Strands.**

Definition 3.3 (Strand graph)**.** Let $\mathcal{T}$ be a homochiral tiling. The *strand graph* $\mathcal{G}(\mathcal{T})$ is the multigraph with one vertex for each piece of each tile and one edge for each pair of ports joined by the straight-through rule, between the two pieces having these ports as ends. A *strand* is a connected component of $\mathcal{G}(\mathcal{T})$; a strand is *closed* if it is finite.

Lemma 3.4. *Every vertex of $\mathcal{G}(\mathcal{T})$ has degree 2. Hence every strand is either a finite cycle (a closed strand) or a two-way infinite path.*

Proof. A piece has two ends, each a port of its tile on some edge e. In a tiling of the plane, e is shared by exactly one other tile, so each end is joined to exactly one port, an end of a piece of that neighbour. So every vertex has degree exactly 2; joins always link pieces of different tiles, so $\mathcal{G}(\mathcal{T})$ has no loops. A connected, locally finite multigraph with all degrees 2 is a finite cycle (possibly of length 2) or a two-way infinite path. □

Equivalently, in a tiling of the plane the ends of the pieces and the straight-through rule pair the ports of all the tiles by two involutions without fixed points, and the ports of a

strand form an orbit of the group they generate. This is how the formalisation describes strands.

4. Boxes and the signature

4.1. **Patches and boxes.** A *patch* is a finite set P of tiles of a homochiral tiling whose union is a closed disc with a simple polygonal boundary, the *rim* ∂P. Every supertile of level ≥ 1 is a patch (Theorem 2.1), and so is a tile. Every rim edge belongs to exactly one tile of P. Let $E = E(P)$ be the number of rim edges, and number the rim vertices $u_0, \dots, u_{E-1}$ in a chosen orientation of the rim, indices mod E; for a supertile we always use its own orientation. For a rim vertex u_k define the *rim ports*

$$\mathrm{R}(u_k) = \text{the port on } u_k u_{k+1} \text{ near } u_k, \qquad \mathrm{L}(u_k) = \text{the port on } u_{k-1} u_k \text{ near } u_k,$$

where the port belongs to the tile of P containing that edge. There are $2E$ rim ports. For the tile itself, with its counter-clockwise orientation, these are the ports $\mathrm{R}(v_i), \mathrm{L}(v_i)$ of Section 3. Reversing the orientation exchanges R and L at every vertex.

Lemma 4.1 (The box of a patch)**.** *Start at a rim port of P and follow the pieces of the tiles of P, crossing every interior edge of P by the straight-through rule. The path ends at a different rim port. This defines an involution π_P on the rim ports without fixed points. The pieces of P that are not reached from any rim port form finitely many cycles, the* closed strands *of P.*

Proof. The pieces of P and the joins across interior edges form a finite graph in which every piece has degree 2, except that a piece end at a rim port has no partner. Its components are therefore paths, whose two ends are rim ports, and cycles. Paths have two distinct ends. □

We call the paths of Lemma 4.1 the *arcs* of P; there are exactly E of them. The rim of P with the pairing π_P is the *box* of P. Everything we need about the inside of P is in its box, together with whether P has a closed strand:

Lemma 4.2 (Gluing)**.** *Let a patch P be the union of patches $P_1, \dots, P_n$ with pairwise disjoint interiors. Then the box of P is determined by the boxes of the P_i together with the identification of their shared rim edges. More precisely, π_P is computed by alternately applying the π_{P_i} and the straight-through rule across shared edges, until a rim port of P is reached. The closed strands of P are those of the P_i together with the cycles of this alternation that contain no rim port of P.*

Proof. Every arc or closed strand of P (Lemma 4.1) is a concatenation of arcs and closed strands of the P_i, joined across edges shared by two of the P_i. □

We use Lemma 4.2 in two ways: to define the box of a composition, glued from boxes alone (Section 7.1), and to compute the box of a supertile from those of its children.

4.2. **The signature.**

Definition 4.3 (Signature)**.** Let A and B be adjacent rim vertices of a patch P. We say that P has the *signature at* (A, B) if

(i) $\pi_P(\mathrm{L}(u)) = \mathrm{R}(u)$ for every rim vertex $u \notin \{A, B\}$ (a *hook* at u);
(ii) $\pi_P(\mathrm{R}(A)) = \mathrm{R}(B)$ and $\pi_P(\mathrm{L}(A)) = \mathrm{L}(B)$ (the *crossover*);
(iii) P has no closed strand: the ribbon followed from any port of P reaches the rim.

The definition is symmetric in A and B and does not depend on the orientation of the rim, since reversing it exchanges L and R. A patch with the signature has $E - 2$ hook arcs, one at each rim vertex other than A and B, and two crossover arcs. Figure 5 is a picture.

Lemma 4.4. *A tile has the signature at (v_0, v_1).*

Proof. A tile has no interior edge, so π is the pairing of Definition 3.1, and every piece has both ends on the rim. □

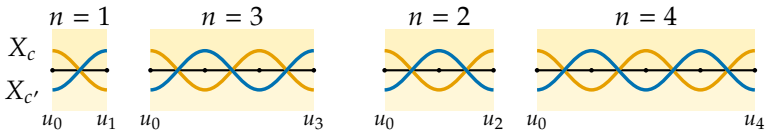


FIGURE 6. The Zigzag Lemma. Along a shared path whose interior corners are hooks on both sides, the ribbons cross the path at every edge and turn back at every corner, alternately in X_c (above) and $X_{c'}$ (below). For odd n each connection ends on the other side; for even n it returns to the side it started from. Only the parity of n matters, so lengthening a shared path by two edges changes nothing (Lemma 7.1).

5. The Zigzag Lemma

In this section a patch P is the union of patches (*children*) $X_0, \dots, X_{n-1}$ with disjoint interiors, each X_c having the signature at some (A_c, B_c). All are oriented in the same rotational sense, so two children traverse a common rim path in opposite directions; $\mathrm{R}_c(u), \mathrm{L}_c(u)$ are the rim ports of X_c at u in this orientation. (For a supertile and its children in their own orientation, P is oriented opposite to its own orientation, which exchanges L and R at the parent's rim vertices but does not affect the signature.)

Lemma 5.1 (Zigzag Lemma)**.** *Let $u_0u_1 \cdots u_n$ ($n \geq 1$) be a path of n edges lying on the rims of both X_c and $X_{c'}$, traversed by X_c from u_0 to u_n. Suppose that none of the interior vertices $u_1, \dots, u_{n-1}$ is a crossover corner of X_c or of $X_{c'}$. Then, alternately applying the straight-through rule across the edges u_ku_{k+1} and the hooks of X_c and $X_{c'}$ at $u_1, \dots, u_{n-1}$, the ports at the two ends of the path are joined as follows:*

$$\begin{aligned} &n \text{ even:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{L}_c(u_n), && \mathrm{R}_{c'}(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0); \\ &n \text{ odd:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{R}_{c'}(u_n), && \mathrm{L}_c(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0). \end{aligned}$$

These two connections use every port on the edges of the path, so no closed strand lies along the path.

Proof. Start at $\mathrm{R}_c(u_0)$, the port of X_c on u_0u_1 near u_0. The straight-through rule takes it to the port of $X_{c'}$ on u_0u_1 near u_1. In the orientation of $X_{c'}$, which runs from u_1 to u_0 along this edge, that port is $\mathrm{R}_{c'}(u_1)$. If $n = 1$ we have reached $\mathrm{R}_{c'}(u_n)$. Otherwise u_1 is a hook of $X_{c'}$, which leads to $\mathrm{L}_{c'}(u_1)$, the port of $X_{c'}$ on u_1u_2 near u_1. Crossing u_1u_2 leads to the port of X_c on u_1u_2 near u_2, which is $\mathrm{L}_c(u_2)$. If $n = 2$ we stop. Otherwise the hook of X_c at u_2 leads to $\mathrm{R}_c(u_2)$, and the pattern repeats with all indices shifted by two. So the path visits the hooks of $X_{c'}$ at odd k and the hooks of X_c at even k. It ends at $\mathrm{L}_c(u_n)$ if n is even and at $\mathrm{R}_{c'}(u_n)$ if n is odd. The same argument from the other end, with X_c and $X_{c'}$ exchanged (and, for odd n, L and R), starting at $\mathrm{R}_{c'}(u_n)$ for even n and at $\mathrm{L}_c(u_n)$ for odd n, gives the second connection; it visits the remaining hooks. Each connection crosses each of the n edges once, using two of its four ports, and the two connections use different ports, so together they use all of them. □

6. Side lengths

6.1. **Letters.** Every side of a level-m supertile ($m \geq 1$) carries one of the edge labels of [29, Fig. 4.2] (Table 4). Ignoring the signs, there are eight *letters*, $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$. Two sides that are glued to each other carry labels x^+ and x^- (or η and η), and so the same letter; this is the matching rule of [29, §4], and for sides glued inside a supertile it can be read off Tables 1, 2 and 4.

kind	side 0	side 1	side 2	side 3	side 4	side 5
Γ	α^-	α^+	γ^-	δ^-	β^+	β^-
Δ	γ^+	β^+	ϵ^-	α^+	γ^-	ζ^-
Θ	γ^+	β^+	θ^+	β^+	η	β^-
Λ	γ^+	β^+	ϵ^-	α^+	θ^-	β^-
Ξ	α^-	ϵ^+	θ^+	β^+	η	β^-
Π	α^-	ϵ^+	ϵ^-	α^+	θ^-	β^-
Σ	ζ^+	β^+	ϵ^-	α^+	γ^-	δ^+
Φ	γ^+	β^+	ϵ^-	ϵ^+	η	β^-
Ψ	α^-	ϵ^+	ϵ^-	ϵ^+	η	β^-

TABLE 4. The label of [29, Fig. 4.2] carried by side j of a supertile of each kind: the label of the edge of the hexagon in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$.

Proposition 6.1 (Lengths by letter)**.** *For every $m \geq 1$ the length $\ell_m(K, j)$ depends only on the letter of side j of K. Writing x_m for the common length of the sides with letter x,*

$$
\begin{aligned}
\alpha_{m+1} &= \beta_m + \epsilon_m + \eta_m, & \beta_{m+1} &= \alpha_m + \epsilon_m,\\
\gamma_{m+1} &= \beta_m + 2\epsilon_m + \eta_m + \theta_m, & \delta_{m+1} &= \alpha_m + \beta_m + \epsilon_m + 2\theta_m,\\
\epsilon_{m+1} &= 3\epsilon_m + \eta_m, & \zeta_{m+1} &= \alpha_m + \epsilon_m + \theta_m,\\
\eta_{m+1} &= 4\epsilon_m + \eta_m, & \theta_{m+1} &= 4\epsilon_m + \eta_m + \theta_m,
\end{aligned}
\tag{1}
$$

with $(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \zeta_1, \eta_1, \theta_1) = (7, 5, 10, 10, 8, 6, 10, 11)$.

Proof. For $m = 1$ this is a direct count on the level-1 supertiles (Fig. 3; Appendix A). By Proposition 2.4(c), $\ell_{m+1}(T, J)$ is the sum of the lengths of the children's sides listed in Table 3. Inspection of Tables 1, 3 and 4 shows that the multiset of letters of these children's sides depends only on the letter of side J of T, and is the one in (1). For example, side 0 of Δ (letter γ) is made of 7.5, 6.4, 6.3, 6.2, 5.4, with letters $\beta, \eta, \epsilon, \epsilon, \theta$. The claim follows by induction on m. □

The recurrence (1) has a closed form. Let F_n be the Fibonacci numbers, $F_1 = F_2 = 1$.

Proposition 6.2 (Fibonacci side lengths)**.** *For every $m \geq 1$,*

$$
\alpha_m = F_{3m+2} + 2, \qquad \beta_m = F_{3m+1} + 2, \qquad \gamma_m = \delta_m = \eta_m = 2F_{3m+2},
$$
$$
\epsilon_m = F_{3m+3}, \qquad \zeta_m = 2F_{3m+1}, \qquad \theta_m = F_{3m+4} - 2.
$$

Proof. For $m = 1$, $(F_4, F_5, F_6, F_7) = (3, 5, 8, 13)$ gives the initial values. For the induction step put $a = F_{3m+1}$ and $b = F_{3m+2}$, so that $F_{3m+3} = a + b$, $F_{3m+4} = a + 2b$, $F_{3m+5} = 2a + 3b$, $F_{3m+6} = 3a + 5b$ and $F_{3m+7} = 5a + 8b$. Substituting into (1):

$$
\begin{aligned}
\alpha_{m+1} &= (a+2) + (a+b) + 2b = F_{3m+5} + 2, & \beta_{m+1} &= (b+2) + (a+b) = F_{3m+4} + 2,\\
\gamma_{m+1} &= (a+2) + 2(a+b) + 2b + (a+2b-2) = 4a + 6b = 2F_{3m+5},\\
\delta_{m+1} &= (b+2) + (a+2) + (a+b) + 2(a+2b-2) = 4a + 6b,\\
\eta_{m+1} &= 4(a+b) + 2b = 4a + 6b, & \epsilon_{m+1} &= 3(a+b) + 2b = F_{3m+6},\\
\zeta_{m+1} &= (b+2) + (a+b) + (a+2b-2) = 2F_{3m+4},\\
\theta_{m+1} &= 4(a+b) + 2b + (a+2b-2) = 5a + 8b - 2 = F_{3m+7} - 2.
\end{aligned}
$$

□

Since F_n is even exactly when $3 \mid n$:

Corollary 6.3 (Parities are constant)**.** *For every $m \geq 1$ the lengths α_m, β_m, θ_m are odd and γ_m, δ_m, ϵ_m, ζ_m, η_m are even.*

6.2. **The crossover corner.**

Lemma 6.4 (Position of A)**.** *Let $m \geq 1$. For $K \neq \Sigma$ the crossover corner A of a level-m K-supertile is its vertex 0. For $K = \Sigma$ it lies on side 5 (letter δ), at distance*

$$d_m = 2F_{3m+1}$$

from vertex 5 and at distance $\delta_m - d_m = 2F_{3m}$ from vertex 0.

Proof. For $K \neq \Sigma$ this is Proposition 2.4(d) for $m \geq 2$, and is read off Fig. 3 for $m = 1$. For $m = 1$, $d_1 = 6$ is read off the level-1 Σ-supertile. By Proposition 2.4(d) and Table 3, the crossover corner of a level-$(m+1)$ Σ-supertile follows the children's sides 0.2, 0.1, 7.0 on its side 5. Child 0 is a Ξ and child 7 a Γ, so these sides have letters θ, ϵ, α and $d_{m+1} = \theta_m + \epsilon_m + \alpha_m = (a+2b-2)+(a+b)+(b+2) = 2F_{3m+4}$, in the notation of the previous proof. □

Lemma 6.5 (Lower bounds)**.** *For every $m \geq 1$: every side has length at least 5; $d_m \geq 6$; and $\delta_m - d_m \geq 4$.*

Proof. By Proposition 6.2 all letter lengths increase with m, and at $m = 1$ the smallest is $\beta_1 = 5$. Also $d_m \geq d_1 = 6$ and $\delta_m - d_m = 2F_{3m} \geq 2F_3 = 4$. □

7. The signature at every level

We prove the Signature Theorem by induction on the level, reducing every step to a finite computation at level 1.

7.1. **Compositions as a function of side lengths.** Fix a parent kind T, with children $c = 0, \ldots, n_T - 1$ of kinds K_c (Table 1), one of which, c_Σ, is a Σ. Let L assign a positive integer $L(c, j)$ to each child c and side $j \in \{0, \ldots, 5\}$, equal on sides glued by Table 2; let d be an integer with $2 \leq d \leq L(c_\Sigma, 5) - 2$, and $\omega \in \{\pm 1\}$. The *composition* $C_T(L, d, \omega)$ is the following abstract patch of boxes. Child c has a rim of $E_c = \sum_j L(c, j)$ vertices, numbered $0, \ldots, E_c - 1$ in its own order, with side j running from vertex $V_c(j) = \sum_{i<j} L(c, i)$ to vertex $V_c(j+1)$; its crossover corner A_c is vertex 0, or for $c = c_\Sigma$ the vertex at distance d from vertex 5 along side 5; $B_c = A_c + \omega$; and it is a box with the signature at (A_c, B_c). Sides are glued in pairs as in Table 2, reversing direction: the o-th edge of side j of c is the o-th edge from the end of side j' of c'. The sides marked ∂ form the rim of the composition, in the order of Table 3 read backwards (the table lists them in the parent's own order); the ports of a rim vertex are those of the children on its two rim edges. Following ribbons alternately through a child, by its signature, and straight across a glued edge, as in Lemma 4.2, defines the box of the composition. The Zigzag Lemma and its proof apply unchanged to compositions, since they use only the signatures of the children and the straight-through rule. We say that $C_T(L, d, \omega)$ *has the signature* if it has no closed strand, and every rim vertex is a hook except the crossover corners A_Γ, B_Γ of its Γ child, which are adjacent rim vertices (B_Γ following A_Γ in the children's order if $\omega = +1$, and preceding it if $\omega = -1$) and carry the crossover.

Write ℓ_m also for the function $(c, j) \mapsto \ell_m(K_c, j)$. If the children of a level-$(m+1)$ T-supertile have the signature at $(A_c, A_c + \omega_m)$ in their own orientation, then the supertile is glued from them as $C_T(\ell_m, d_m, \omega_m)$ is. Indeed: (i) by Proposition 2.4(a),(b) and Proposition 6.1 the children have the kinds and side lengths of the composition, and side j of X_c is side j' of $X_{c'}$ traversed oppositely; (ii) the rims of two children meet only in the closed sides they share: at a point inside a side both rim edges are shared with the partner, so the two children fill a neighbourhood, and a vertex lies on exactly three level-m supertiles, any two of which share a side ending there (Theorem 2.1(b)); (iii) so each side of a child is wholly shared with one sibling or wholly on the rim, as in Table 2; (iv) A_c is placed by Lemma 6.4, and $B_c = A_c + \omega_m$ by hypothesis; (v) the rim of the parent is given by Table 3 (Proposition 2.4(c)). By Lemma 4.2, the strands of the supertile are then those of the composition, and its box is the box of $C_T(\ell_m, d_m, \omega_m)$. Step (ii) is the one geometric step; with Theorem 2.1 and Proposition 2.4 it

is part of the assumption (M1) of Appendix B. The compositions themselves are built from the tables alone, which agree with each other: Table 3 lists each side that Table 2 puts on the rim exactly once, and turning about the end of each entry, across the sides that Table 2 glues, leads to the start of the next (a finite check, made in the formalisation).

7.2. **Lengthening.** Let (L, d) be as above. A *move* lengthens one side of one child by two edges, together with the side glued to it if there is one. An edge e of that side is replaced by a path of three edges through two new vertices; e must not be the first or the last edge of its side (in each child that contains it), nor one of the two edges at A_Σ (the crossover corner of the Σ child). The result is (L', d'), where L' is L with the side (and its partner) two edges longer, and $d' = d + 2$ if e lies on side 5 of the Σ child before A_Σ, and $d' = d$ otherwise.

Lemma 7.1 (Lengthening)**.** *If $C_T(L, d, \omega)$ has the signature, then so does $C_T(L', d', \omega)$, for every move $(L, d) \to (L', d')$.*

Proof. Write $C = C_T(L, d, \omega)$ and $C' = C_T(L', d', \omega)$, and let $e = u_0u_1$. Then C' is C with e cut into three: in each child that contains it, e is replaced by the path $u_0v_1v_2u_1$, and every vertex of C has an image vertex in C' (positions after e move on by two). The conditions on e make the corners of C' the images of those of C: for A_Σ by the choice of d', and $B_c = A_c \pm 1$ stays next to A_c because the edge between them (the first edge of side 0, the last of side 5, or an edge at A_Σ) is never cut. In particular v_1 and v_2 are not crossover corners. Let φ map each port of C to the port of C' of the same hand at the image vertex. Then φ commutes with the pairings inside the children, since the corners correspond, and with the straight-through rule across every glued edge other than e. If e is shared by children c and c', crossing it joins $\mathrm{R}_c(u_0)$ to $\mathrm{R}_{c'}(u_1)$ and $\mathrm{L}_c(u_1)$ to $\mathrm{L}_{c'}(u_0)$ (Lemma 5.1 with $n = 1$); in C', by Lemma 5.1 with $n = 3$, the ribbons run along the new path from $\varphi(\mathrm{R}_c(u_0))$ to $\varphi(\mathrm{R}_{c'}(u_1))$ and from $\varphi(\mathrm{L}_c(u_1))$ to $\varphi(\mathrm{L}_{c'}(u_0))$. If e lies on the rim, v_1 and v_2 are hooks of the child, and so new rim vertices of C' that are hooks of C'.

Hence, for every port x of C, the ribbon of C' from $\varphi(x)$ passes through the images of the ports met from x in C, and reaches the rim at the image of the rim port reached in C. So C' has a hook at the image of every rim vertex of C except A_Γ and B_Γ, and a hook at each new rim vertex; A_Γ and B_Γ are adjacent as before, are still rim vertices, since the rim edges of the Γ child at them stay on the rim, and carry the crossover. Finally, C' has no closed strand: every port of C' at an old vertex is some $\varphi(x)$, whose ribbon reaches the rim; and a strand through a port at v_1 or v_2 is a hook of the rim, or follows one of the two connections of Lemma 5.1 along $u_0v_1v_2u_1$, which use every port on these edges and end at ports at u_0 and u_1, so it too reaches the rim. □

Call (L, d) *admissible* if sides glued by Table 2 have the same length; $L(c, j) \geq \ell_1(c, j)$ and $L(c, j) \equiv \ell_1(c, j) \pmod 2$ for all c and j; $d \geq d_1$ and $d \equiv d_1 \pmod 2$; and $L(c_\Sigma, 5) - d \geq \ell_1(c_\Sigma, 5) - d_1 = 4$. Then $d \geq 6$ and $L(c_\Sigma, 5) - d \geq 4$, so the composition $C_T(L, d, \omega)$ is defined.

Lemma 7.2 (Reachability)**.** (a) *For every $m \geq 1$, the pair (ℓ_m, d_m) is admissible.*

(b) *Every admissible $(L, d) \neq (\ell_1, d_1)$ arises by a move from an admissible pair (L_0, d_0) with $\sum_{c,j} L_0(c, j) < \sum_{c,j} L(c, j)$. The move can be chosen so that its edge e lies at least two edges from both ends of its side, in each child that contains it, unless that side is side 5 of the Σ child or is glued to it.*

Hence a property of pairs that holds at (ℓ_1, d_1) and is kept by the moves of (b) *holds at every admissible pair, in particular at (ℓ_m, d_m) for every $m \geq 1$.*

Proof. (a) Glued sides have the same letter (Section 6), so the same length. By Proposition 6.2 the letter lengths do not decrease with m, and by Corollary 6.3 their parities do not change. By Lemma 6.4, $d_m = 2F_{3m+1} \geq 6 = d_1$ is even, and $\ell_m(c_\Sigma, 5) - d_m = \delta_m - d_m = 2F_{3m} \geq 4$.

(b) If $L = \ell_1$, then $L(c_\Sigma, 5) = \delta_1 = 10$ and $d_1 \leq d \leq 10 - 4$, so $d = d_1$. Hence $L \neq \ell_1$, and by the parity condition some side j of some child c has $L(c, j) \geq \ell_1(c, j) + 2$; we undo a move on that side. If the side is neither side 5 of Σ nor glued to it, take $d_0 = d$ and the move whose

edge is the third edge of the side (the third from the end in the partner): before the move the side has length at least $\ell_1(c, j) \geq 5$ (Lemma 6.5), so this edge lies at least two edges from both ends, in either child. Otherwise, measure positions along side 5 of the Σ child from its vertex 5. If $d \geq d_1 + 2$, take $d_0 = d - 2$ and the move whose edge is the second edge of the side, before A_Σ; it does not touch A_Σ, since $d_0 \geq 6$. If $d = d_1$, take $d_0 = d$ and the second edge after A_Σ: before the move the side has $L(c_\Sigma, 5) - 2 - d \geq \ell_1(c_\Sigma, 5) - d_1 = 4$ edges after A_Σ, so this edge is at least two edges from the end of the side. In each case (L_0, d_0) is admissible, and $\sum L_0 = \sum L - 2$ or $\sum L - 4$. The last statement follows by induction on $\sum_{c,j} L(c, j)$. □

Computation 7.3 (The compositions at the level-1 lengths)**.** For each of the nine kinds T and both signs ω, the composition $C_T(\ell_1, d_1, \omega)$ has the signature.

This was computed by a program that builds the 18 compositions from Tables 1 to 3 and the lengths alone and follows the ribbon from every port, and again in the kernel of Lean (Appendix A).

Proposition 7.4 (Compositions at every level)**.** *For every $m \geq 1$, every kind T and both signs ω, the composition $C_T(\ell_m, d_m, \omega)$ has the signature.*

Proof. Fix T and ω. The property that $C_T(L, d, \omega)$ has the signature holds at (ℓ_1, d_1) by Computation 7.3, and is kept by every move by Lemma 7.1. By Lemma 7.2 it holds at (ℓ_m, d_m). □

7.3. **The induction.**

Computation 7.5 (Base case)**.** Every level-1 supertile has the signature at (A, B), where A is as in Lemma 6.4 and $B = A + \omega_1$ in own order. Moreover ℓ_1 and d_1 are as stated in Proposition 6.1 and Lemma 6.4.

This was computed from the tiles in exact arithmetic, with Lemma 4.2, checking the statement as worded, and again by a second program and in the kernel of Lean (Appendix A).

Theorem 7.6 (Signature Theorem)**.** *In every homochiral tiling by* Tile$(1, 1)$*, for every $m \geq 1$, every level-m supertile of every kind has the signature at (A, B), where A is its crossover corner (Section 2.3), placed as in Lemma 6.4, and $B = A + \omega_m$ in its own order. In particular no such supertile contains a closed strand.*

Proof. For $m = 1$ this is Computation 7.5. Assume the theorem for level $m \geq 1$ and let Y be a level-$(m + 1)$ supertile of kind T. Its children have the signature at $(A_c, A_c + \omega_m)$, so by Section 7.1 (with Lemma 4.2) its box is that of $C_T(\ell_m, d_m, \omega_m)$, which has the signature by Proposition 7.4. Hence Y has no closed strand, and hooks at all its rim vertices except the adjacent crossover corners A_Γ, B_Γ of its Γ child, which carry the crossover. By definition A_Γ is the crossover corner of Y, placed as in Lemma 6.4. Finally $B_\Gamma = A_\Gamma + \omega_m$ in the child's own order, which is opposite to that of Y; so $B = A - \omega_m = A + \omega_{m+1}$ in the own order of Y. □

7.4. **No closed strands.**

Proof of Theorem A. Let $\mathcal{T}$ be a homochiral tiling whose hierarchy covers the plane: for some tile t, the supertiles $X_0(t) \subset X_1(t) \subset \cdots$ containing t have union $\mathbb{R}^2$. Write $X_m = X_m(t)$. First, about points of the plane: every tile t' lies in some X_m, and every edge of t' is shared by two tiles of some X_m. Indeed, a point x of the interior of t' lies in X_m for some m, hence in some tile s of X_m. Every neighbourhood of x meets the interior of s, so the interiors of s and t' meet, and $s = t'$. An edge of t' is shared with exactly one other tile (Lemma 3.4), and the two tiles lie in some X_m.

The rest is combinatorial. Let $X_1 \subset X_2 \subset \cdots$ be supertiles, each a child of the next, such that every edge of every tile of their union is shared by two tiles of some X_m; we show that every strand of the union is a two-way infinite path. A closed strand has finitely many pieces, in finitely many tiles, and is joined across finitely many edges, so these all lie in a

single X_m, each edge shared by two of its tiles. It is then a closed strand of X_m, contradicting Theorem 7.6. By the argument of Lemma 3.4, every strand is a two-way infinite path. This second step is Theorem A as the formalisation states it (Appendix B). □

8. The nest tiling

Child 1 of every supertile is a Δ (Table 1).

Construction 8.1 (The nest)**.** Let $\mathcal{T}$ be a homochiral tiling [29, Thm. 2.2]. For each $h \geq 0$, let $Y^{(h)}$ be a level-h Δ-supertile of $\mathcal{T}$; one exists, for instance child 1 of any level-$(h+1)$ supertile. Let $Y^{(h)} = Y^{(h)}_h \supset Y^{(h)}_{h-1} \supset \cdots \supset Y^{(h)}_0$ be its chain of descendants, each child 1 of the next. Let g_h be the proper motion taking the tile $Y^{(h)}_0$ onto Tile$(1,1)$ (with v_0 at the origin), and put $N^{(h)}_b = g_h(Y^{(h)}_b)$ for $b \leq h$.

Proposition 8.2. *$N^{(h)}_b$ does not depend on $h \geq b$ (or on the choices made). Writing N_b for it, $N_0 = \{\text{Tile}(1,1)\}$, and N_b is child 1 of N_{b+1} for every b.*

Proof. Let $h, h' \geq b$, with $Y^{(h)}$, $Y^{(h')}$, g_h, $g_{h'}$ as in Construction 8.1. By Lemma 2.6 there is a proper motion f with $f(Y^{(h)}_b) = Y^{(h')}_b$ that maps descendants to descendants with their numbers; hence $f(Y^{(h)}_{b'}) = Y^{(h')}_{b'}$ for all $b' \leq b$, in particular for the tiles Y_0. So $g_{h'}f$ and g_h both take the tile $Y^{(h)}_0$ onto Tile$(1,1)$, and since a tile has no symmetry they are equal. Thus $N^{(h')}_b = g_{h'}f(Y^{(h)}_b) = N^{(h)}_b$. □

The *nest tiling* is $\mathcal{N} = \bigcup_{b\geq 0} N_b$, a set of tiles.

Lemma 8.3 (Grandparent Lemma)**.** *Let $b \geq 1$.* (a) *The block N_b does not meet the rim of N_{b+2}.* (b) *Every edge of a tile of N_b is shared by two tiles of N_{b+2}.*

Proof. (a) N_b is child 1 of N_{b+1} (Proposition 8.2). By Table 2, the sides $0, 1, 3, 4, 5$ of child 1 face the siblings $7, 0, 2, 3, 4$, and side 2 lies on the parent's rim. A point of N_b that is not on its closed side 2 is therefore an interior point of N_{b+1}. This is clear for interior points of N_b and for interior points of the other sides, where N_b and one sibling together fill a neighbourhood. At the vertices $0, 1, 4, 5$ of N_b, exactly three level-b supertiles meet; two of them are the distinct siblings across the adjacent sides, and together they fill a neighbourhood. It remains to treat the closed side 2 of N_b. By Table 3 (the Δ row: positions 14 to 18), side 4 of the parent is the concatenation of 2.2, 2.1, 1.2, 0.4, 0.3. Since 1.2 is neither first nor last, the closed side 2 of N_b lies in the relative interior of side 4 of N_{b+1}. In N_{b+2}, side 4 of its child N_{b+1} faces the sibling 3, so, as above one level up, the points of its relative interior are interior points of N_{b+2}.

(b) A rim edge of N_b lies on a side $j \neq 2$ of N_b, which faces a sibling inside N_{b+1}, or on side 2, which lies on side 4 of N_{b+1} (position 16 of Table 3); and side 4 of child 1 faces sibling 3 inside N_{b+2} (Table 2). Any other edge of a tile of N_b is shared by two tiles of N_b. □

Proposition 8.4. *$\mathcal{N}$ is a homochiral tiling of the plane, and every tile of $\mathcal{N}$ lies in some N_b.*

Proof. Any two tiles of $\mathcal{N}$ lie in a common block, so their interiors are disjoint. We show that the tiles cover the plane. Fix h and work in the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$ of Construction 8.1, which contains $N_0, \ldots, N_h$ (Proposition 8.2). Call two tiles of $\mathcal{T}_h$ *neighbours* if they share a point, and let dist be the resulting graph distance. We claim that every tile at distance at most k from N_1 lies in N_{1+2k}, provided $1 + 2k \leq h$. For $k = 0$ there is nothing to prove. If t has distance $k+1$, it shares a point p with a tile of N_{1+2k}. By Lemma 8.3, p is an interior point of N_{3+2k}, so the interior of t meets the interior of a tile of N_{3+2k}; hence t is that tile. Now let $x \in \mathbb{R}^2$ and let D be the disc of radius $r = |x| + 1$ about the origin. The tiles of $\mathcal{T}_h$ meeting D have diameter $3 + \sqrt{3} < 5$ and area $3 + 3\sqrt{3} > 8$, so there are at most $k_0 = \lfloor \pi(r+5)^2/8 \rfloor$ of them. They cover the connected set D, so their neighbour graph is connected. One of them is $N_0 \subset N_1$, since the origin is a vertex of N_0. So every tile of $\mathcal{T}_h$

b	tiles in N_b	arcs of N_b	$k=1$	$k=2$	$k=3$	$k=4$
0	1	14	5	2	1	1
1	9	46	10	2	2	1
2	71	182	37	2	1	1
3	559	758	146	2	2	1
4	4,401	3,198	613	2	1	1
5	34,649	13,534	2,586	2	2	1
6	272,791	57,318	10,949	2	1	–
even $b \geq 2$				2	1	1
odd $b \geq 1$				2	2	1

TABLE 5. The number of arcs of N_{b+k} on which the arcs of N_b lie, that is, the number of blocks of $\mathcal{P}_k(b)$ (row $b = 0$: the 14 pieces of the single tile, computed on the drawing up to level 5). Rows $1 \leq b \leq 6$: computed by following every ribbon, up to level 9 (in floating point; for $b \leq 3$ also in exact arithmetic, up to level 7). Last two rows: the proof of Theorem 9.1, for every $b \geq 1$ of each parity.

meeting D, in particular one containing x, lies in N_{1+2k_0} as soon as $h \geq 1 + 2k_0$. Since N_{1+2k_0} does not depend on h, x is covered by $\mathcal{N}$. □

The blocks N_b are supertiles in the sense of Section 2: each is congruent, with all its structure, to a level-b Δ-supertile of $\mathcal{T}$. So Theorem 7.6 applies to them. In particular $\mathcal{N}$ has no closed strand: a closed strand passes through finitely many tiles, which lie in a common block N_b, $b \geq 1$, by Proposition 8.4; by Lemma 8.3(b) every edge of these tiles is shared by two tiles of N_{b+2}, so it would be a closed strand of N_{b+2}.

9. Merging and the single strand

9.1. **Merging.** For $b \geq 1$ and $k \geq 0$, every arc of N_b is contained in a unique arc of N_{b+k}, because N_{b+k} has no closed strand. Let $\mathcal{P}_k(b)$ be the partition of the $E_b(\Delta)$ arcs of N_b in which two arcs are in the same block when they lie on the same arc of N_{b+k}. The blocks only merge as k increases; Table 5 gives their numbers.

Theorem 9.1 (Merging)**.** *For every $b \geq 1$, $\mathcal{P}_4(b)$ has a single block; if b is even, so does $\mathcal{P}_3(b)$. These bounds are exact: $\mathcal{P}_2(b)$ has two blocks, and so does $\mathcal{P}_3(b)$ if b is odd.*

Junctions. Let $m \geq 1$. The rim of N_{m+1}, in its own order from its corner A, is the concatenation of the 22 children's sides listed in Table 3 for eight children, at positions $q = 0, \ldots, 21$, with its sides starting where the Δ row says. The *junction* q is the rim vertex of N_{m+1} at which the side at position q starts. The junctions $9, 12, 16, 17, 21$ lie inside sides of N_{m+1} (Lemma 9.4 below), so they are hooks of N_{m+1} (Theorem 7.6); we write H_q for the hook arc of N_{m+1} at junction q. The side at position 16 is 1.2, side 2 of N_m, so junctions 16 and 17 are its two ends: vertices 3 and 2 of N_m. Positions on the rim of N_m are taken in its own order from its corner A, so that its vertices $0, \ldots, 5$ lie at positions $0 = V_0 < V_1 < \cdots < V_5 < E$, where $E = E_m(\Delta)$; its corner $B = A + \omega_m$ is at position 1 if m is even and $E - 1$ if m is odd.

Computation 9.2 (The junction map at the level-1 lengths)**.** In $C_\Delta(\ell_1, d_1, \omega)$, for both signs ω, with child 1 in the place of N_m and positions and junctions defined in the same way, the ribbon from every port of child 1, except those at rim vertices strictly inside its side 2, reaches the rim of the composition at junction q, where q is given by Table 6 with $\omega_m = \omega$.

This was computed like Computation 7.3, by a program and in the kernel of Lean.

arc of N_m	m even ($\omega_m = +1$)	m odd ($\omega_m = -1$)
crossover arc through R(A) and R(B)	17	21
crossover arc through L(A) and L(B)	9	16
hook at u, $0 < u \leq V_1$	17	21
hook at u, $V_1 < u \leq V_2$	17	17
hook at u, $V_2 < u < V_3$	a hook of N_{m+1} itself	
hook at u, $V_3 \leq u < V_4$	16	16
hook at u, $V_4 \leq u < V_5$	12	16
hook at u, $V_5 \leq u < E$	9	16

TABLE 6. The junction map (Lemma 9.3): the junction q such that an arc of N_m lies on the hook H_q of N_{m+1}. Hooks are listed by the position u of their rim vertex, which is never A or B.

Lemma 9.3 (The junction map)**.** *Let $m \geq 1$. In N_{m+1}, every hook of N_m at a rim vertex strictly inside side 2 of N_m is a hook of N_{m+1}, and every other arc of N_m lies on the hook H_q of N_{m+1}, where q is given by Table 6.*

Proof. The first part needs no computation. Side 2 of N_m is the side at position 16 of Table 3, so it lies on the rim of N_{m+1}, and a rim vertex u of N_m strictly inside it is a rim vertex of N_{m+1}, with the same two rim ports. By Lemma 9.4, u lies inside side 4 of N_{m+1}, so it is not a crossover corner, and by Theorem 7.6 it is a hook of N_{m+1}; this hook is the hook of N_m at u.

For the rest, say that (L, d) has the *junction property* if the statement of Computation 9.2 holds for $C_\Delta(L, d, \omega)$ for both ω. By Theorem 7.6 and Section 7.1, N_{m+1} is glued from its children as $C_\Delta(\ell_m, d_m, \omega_m)$ is, with N_m as child 1; a ribbon of the composition from a port p of child 1 runs along the arc of N_m ending at p, and on along the arc of N_{m+1} that contains it; and a hook arc has both its ends at one rim vertex. So the rest of the lemma says that (ℓ_m, d_m) has the junction property. By Computation 9.2, (ℓ_1, d_1) has it, so by Lemma 7.2 it suffices to show that the moves of Lemma 7.2(b) keep it. Since child 1 is a Δ, and the side glued to side 5 of the Σ child (child 4) belongs to child 7 (Table 2), such a move whose edge lies on the rim of child 1 keeps it at least two edges from both ends of its side.

Let C, C' and φ be as in the proof of Lemma 7.1. Junctions and the vertices V_j of child 1 are vertices of children, so those of C' are the images of those of C, and φ keeps the order of positions on the rim of child 1. For a port x of child 1 at an old vertex, the ribbon from $\varphi(x)$ reaches the rim at the image of the rim port reached from x, and $\varphi(x)$ lies in the same row of Table 6 as x. It remains to consider the new vertices v_1, v_2 when the edge $e = u_0u_1$ lies on the rim of child 1. Then v_1, v_2, u_0, u_1 lie strictly inside the same side of child 1, at least two edges from its ends, so in the same row of Table 6, and none of them is a crossover corner. If that side is side 2, the property says nothing about v_1 and v_2. Otherwise e is shared with a sibling c'. Write $\mathrm{R}_1, \mathrm{L}_1$ and $\mathrm{R}_{c'}, \mathrm{L}_{c'}$ for the rim ports of the two children, and let child 1 run from u_0 to u_1. From $\mathrm{L}_1(v_2)$, the hook of child 1 at v_2 and one crossing lead to $\mathrm{R}_{c'}(u_1)$, as in C the hook at u_0 and the crossing of e lead there from $\mathrm{L}_1(u_0)$. From $\mathrm{R}_1(v_2)$, the hook of child 1 at v_2, the hook of c' at v_1 and two crossings lead to $\mathrm{R}_1(u_0)$. In both cases the ribbon then runs on as in C, under φ, so it reaches the rim where the ribbon of C from the port of the same hand at u_0 does. Likewise at v_1, with u_1 in place of u_0. Those ports lie in the same row, so the property holds at v_1 and v_2 too. □

Lemma 9.4 (Where the junctions lie)**.** *Let $m \geq 1$. The junctions 9, 12, 16, 17 and 21 of N_{m+1} lie on its sides 2, 3, 4, 4 and 5 respectively, at least two edges from both ends of the side. Every rim vertex strictly inside side 2 of N_m lies on side 4 of N_{m+1}, at least two edges from both ends.*

Proof. By Table 3, side 2 of a Δ is made of the children's sides at positions 7 to 10, side 3 of those at 11 to 13, side 4 of those at 14 to 18, and side 5 of those at 19 to 21. Junction q lies between the sides at positions $q - 1$ and q, which lie in the same side of N_{m+1} for

$q = 9, 12, 16, 17, 21$; and side 2 of N_m is the side at position 16. Every side of a supertile of level ≥ 1 has at least five edges (Lemma 6.5). □

Proof of Theorem 9.1. A rim vertex of N_m at least two edges from both ends of its side J is not a crossover corner, and by Lemma 9.3 its hook lies on the hook of N_{m+1} that Table 6 names for side J: for even m, junction 17, 17, 16, 12, 9 for $J = 0, 1, 3, 4, 5$; for odd m, junction 21, 17, 16, 16, 16; and for $J = 2$, the vertex is itself a hook of N_{m+1}, on its side 4. By Lemma 9.4, each of these hooks of N_{m+1} lies at least two edges inside a side of N_{m+1}, so the same rule applies to it one level up.

Let $b \geq 1$ be even, so that $\omega_b = +1$. By Lemma 9.3 every arc of N_b lies on H_{17}, H_{16}, H_{12} or H_9 of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b; by Lemma 9.4 these are hooks inside sides 4, 4, 3, 2 and 4 of N_{b+1}. Since $b + 1$ is odd, the hooks inside sides 3 and 4 lie on H_{16} of N_{b+2}, and those inside side 2 are hooks of N_{b+2} inside its side 4. All are inside side 4 of N_{b+2}, and since $b + 2$ is even they lie on H_{12} of N_{b+3}. So $\mathcal{P}_3(b)$ has a single block.

Let $b \geq 1$ be odd. Every arc of N_b lies on H_{21}, H_{17} or H_{16} of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b: inside sides 5, 4, 4, 4 of N_{b+1}. Since $b + 1$ is even, those inside side 4 lie on H_{12} and those inside side 5 on H_9 of N_{b+2}, inside its sides 3 and 2. Since $b + 2$ is odd, the first lie on H_{16} of N_{b+3}, and the second are hooks of N_{b+3} inside its side 4. All are inside side 4 of N_{b+3}, and since $b + 3$ is even they lie on H_{12} of N_{b+4}. So $\mathcal{P}_4(b)$ has a single block. Once $\mathcal{P}_k(b)$ has a single block, so does $\mathcal{P}_{k'}(b)$ for every $k' \geq k$.

The same steps give the exactness. For even b the arcs of N_b lie on two arcs of N_{b+2}: H_{16}, and the hook at junction 9 of N_{b+1}, which lies inside side 2 of N_{b+1} and so is not junction 16 of N_{b+2} (an end of that side). For odd b they lie on H_{12} and H_9 of N_{b+2}, and on H_{16} of N_{b+3} and the hook at junction 9 of N_{b+2}, which for the same reason is not junction 16 of N_{b+3}. Both arcs are reached in each case (for instance from vertices well inside sides 4 and 5 of N_b, or sides 4 and 0), and distinct hooks are distinct arcs. □

9.2. **One strand.**

Proof of Theorem B. By Lemma 3.4, every component of the strand graph $\mathcal{G}(\mathcal{N})$ is a cycle or a two-way infinite path, and there are no cycles, as shown after Proposition 8.4. It remains to show that $\mathcal{G}(\mathcal{N})$ is connected. Let p, q be pieces of tiles t, t' of $\mathcal{N}$. By Proposition 8.4 both tiles lie in some N_b with $b \geq 1$. Since N_b has no closed strand, p and q lie on arcs of N_b. By Theorem 9.1 these arcs lie on one arc of N_{b+4}, so p and q are on the same strand. The strand passes through every tile, since every tile has pieces. □

10. Scope and open questions

Theorem B is proved for the nest tiling, whose chain of supertiles is $\Delta \subset \Delta \subset \cdots$ with child number 1 at every step. Every tiling whose hierarchy covers the plane has some chain $(K_m, c_m)_{m \geq 1}$, where X_m is a level-m supertile of kind K_m and child c_m of X_{m+1}. The induction of Section 7 does not see the chain, but the merging of Section 9 does. Computations, not used here, indicate that the outcome depends on it. Along the chain $K_m = \Theta$, $c_m = 2$ (odd m), $K_m = \Gamma$, $c_m = 7$ (even m), built as in Construction 8.1 and covering the plane by the argument of Lemma 8.3 and Proposition 8.4, the arcs of X_1 lie on exactly three arcs of X_m for $3 \leq m \leq 8$ (exact arithmetic), and still on three at $m = 20$ (a symbolic count); we have not proved that this persists. Along 90,000 chains chosen at random, the arcs of X_b ($b \leq 3$) lay on a single arc of X_m for some $m \leq 20$ in all but five (folder `experiments/`). Which tilings whose hierarchy covers the plane carry a single strand, and which numbers of strands occur, is open.

In a tiling with a *fault*, the supertiles containing a tile fill only part of the plane, an *infinite supertile* [31, §5], and two or more of these meet along the fault; Tatham [31, §7.3] gives an example. A closed strand inside one infinite supertile is ruled out by Theorem 7.6: its finitely many tiles lie in one supertile, and every edge it crosses is shared by two of them, so it is a closed strand of that supertile. One crossing the fault is not reached by the argument.

Whether such a strand can occur is open. Theorem B is not affected, since the nest tiling has no fault (Proposition 8.4).

The theorems use only the pairing of ports (Definition 3.1) and the straight-through rule, so they hold for any drawing that realises these, including on the curved Spectre. For another decoration with the signature, the same finite computations can be run; their outcome may differ.

Appendix A. The computations

All programs and their logs are in the public repository `https://github.com/Cortado-J/SPECTRE_KNOT_PROOF`, folder `paper/code/`: `./run_all.sh` re-runs everything the proof needs, and its `README.md` maps every statement of this paper to the script that checks it. The Lean formalisation is in the folder `lean/` (Appendix B). The proof site, `https://cortado-j.github.io/SPECTRE_KNOT_PROOF/`, gives each result with its proof, what it cites and the Lean that formalises it.

What the proof computes. The geometric computations on actual supertiles (the level-1 data and Computation 7.5; the patterns of the compositions $m \to m+1$ for $m \le 3$) are done in exact arithmetic by `exact/exact.py`, with every vertex in $\mathbb{Z}[\rho]$ and every motion $z \mapsto \rho^k z + t$; floating point is used only to read off the sign of an area exceeding 8 in absolute value. The computations on compositions (Computations 7.3 and 9.2) use only Tables 1 to 3 and the integers ℓ_1, d_1: `hierarchy/compositions.py` builds each composition, follows the ribbon from every port and checks the statement as worded. The Lean kernel repeats Computations 7.3, 7.5 and 9.2, and runs two negative controls as tests, which the programs run too: with each child's corner B_c two vertices from A_c, all 18 checks of Computation 7.3 fail; with the junctions 12 and 16 exchanged in Table 6, Computation 9.2 fails for both signs. For Proposition 2.4, a script fits our level-1 supertiles to a render of [29, Fig. 4.1] (the six dots of each cluster by a proper similarity with residual below 1% of an edge, every other matching or kind at least 40 times worse; the outline, the Mystic's shading; the labels, read by hand), and another encodes the nine panels of [29, Fig. 5.1], finds the one consistent rotation of each inner hexagon and checks the encoding against Tables 2 and 3.

Evidence, not proof. Besides Remark 2.5, separate programs, mostly in JavaScript, sharing no code with `exact.py` and working in floating point (we do not bound their rounding error), agree with the proof: the actual supertiles up to level 9 have the signature; following every ribbon through the actual blocks up to level 9 gives the numbers of Table 5, and Table 6 holds on $N_1, \dots, N_4$ (exact arithmetic); and second routes, through *skeletons* of the compositions (only the parities of the shared runs) and a symbolic computation with certified linear inequalities, give Theorems 7.6 and 9.1 again. The drawn ribbons realise the pairing of Definition 3.1 on all supertiles up to level 3; each piece stays inside its tile, the two crossover pieces cross once, transversally, and the pieces meet each edge at $\pm 45°$, so the curves cross only in pairs.

Appendix B. Use of AI, formal verification and tools

How the work was done. The author came up with the original idea of looking for a mono-stranded aperiodic knot, and then used an AI system (Claude, by Anthropic) to search for it; from the 34 candidates found, the author chose design 0.1:01 and drew its ribbons by hand. From then on the author used the AI system at every stage: it did the detailed work (the programs, the proof, the Lean formalisation and this paper), and the author directed, guided and corrected it. The author has read parts of this paper and spot-checked figures and counts; no professional mathematician has reviewed it. A program or check called *independent* or *separate* here is a separate implementation, written with the same AI system. At the author's direction, the AI system also reviewed the argument in several passes and searched the literature for related work. AI systems can reproduce published ideas without

naming their source, so this paper may use an idea whose source it does not cite. A fuller account: https://cortado-j.github.io/SPECTRE_KNOT/made.html.

Formal verification. The combinatorial part of the proof is formalised in the Lean 4 proof assistant [8], using its core library only (folder `lean/`). In a combinatorial form, in which a patch, or a whole tiling, is described by which edge of which tile is glued to which, Lean's kernel has checked Theorem 7.6 for every level $m \geq 1$; Theorem A for the union of any chain of supertiles, each a child of the next, in which every edge is eventually glued (the second step of its proof; the nest is such a chain); Theorem B for the nest tiling built as the union of the blocks N_b, with Theorem 9.1, its exactness included; and Lemma 8.3, part (a) at the vertices of tiles. The formal statements rest on Tables 1 to 3 and on the nine level-1 supertiles as glued patches of tiles, computed in exact arithmetic. The link with tilings of the plane is a stated assumption, in three parts: (M1) the supertiles of an actual tiling are glued as the tables say (Theorem 2.1, Proposition 2.4 and Lemma 2.6, and step (ii) of Section 7.1); (M2) a tiling whose hierarchy covers the plane is the union of such a chain (the first step of the proof of Theorem A); (M3) Propositions 8.2 and 8.4, with the step of Lemma 8.3(a) from the vertices of tiles to the points of the plane. The formal proofs follow Sections 3 to 9 step by step (`lean/README.md` maps each result to its formal counterpart), with differences of bookkeeping only: strands are orbits of ports (Section 3.3); the signature is stated by following ribbons (Definition 4.3); and inside the proofs of Lemma 7.1 and Proposition 7.4 a composition's rim vertices are found by turning about a vertex of a child, which cutting an edge leaves unchanged, with the positions of Table 3 and the adjacency of A and B recovered at the end. The formalisation also proves that a lengthened composition is the one with an edge cut into three, and facts the paper takes from the geometry: every supertile is a disc at its rim, and its gluing is symmetric and glues no tile to itself. The proofs use only Lean's standard axioms. Like the rest of the work, the formalisation was written with the AI system; its statement file (`lean/SpectreKnot/Statement.lean`) contains every definition that the formal theorems depend on, with comments in plain English. No human has checked that these formal statements say what the theorems of this paper say.

Tool and computational resource disclosure. *AI system:* Claude, by Anthropic (various Claude models, primarily Opus), used in the Claude app and through Claude Code from August to October 2026; no other AI or machine-learning system was used. *Proof assistant:* Lean 4, version 4.34.1, core library only; `leanchecker` [21] replays every declaration through the kernel, and `comparator` [20] checks that the main theorems prove exactly the statements of `Statement.lean`. *Other software:* Python 3 (SymPy for Section 6; NumPy and SciPy only to propose certificates, then checked in exact arithmetic; Matplotlib and Pillow for the comparisons with [29]), Node.js and Poppler, in the versions recorded in `logs/A00_environment.log`, and LaTeX; one cross-check downloads Kaplan's `spectre.js`, which is not included. *Computing:* the computations the proof needs take about twelve minutes on a four-core machine, the corroborating ones to level 9 about an hour (up to 6.5 GB of memory), and the formalisation about ten minutes; none of the checks the proof needs uses the AI system or proprietary software. The AI system ran on Anthropic's servers; its computing is not known to the author.

References

1. Michael Baake and Uwe Grimm, *Aperiodic order. Volume 1: A mathematical invitation*, Encyclopedia of Mathematics and its Applications, vol. 149, Cambridge University Press, Cambridge, 2013.
2. George Bain, *Celtic art: The methods of construction*, William MacLellan, Glasgow, 1951.
3. bohemian-miser, *Spectre: playing with the spectre tiling*, https://github.com/bohemian-miser/Spectre, file `docs/FASS_THEOREM.md` at commit `f1a946c` (accessed 8 October 2026), 2026.
4. Joseph I. Cline, *Interlace patterns emerging in a Penrose-type Islamic design*, Proceedings of Bridges 2024: Mathematics, Art, Music, Architecture, Culture, 2024, pp. 163–170.
5. ———, *Knotted strapwork strands in a Penrose-type girih pattern*, Proceedings of Bridges 2025: Mathematics, Art, Music, Architecture, Culture, 2025, pp. 61–68.

6. Peter R. Cromwell, *Celtic knotwork: mathematical art*, The Mathematical Intelligencer **15** (1993), no. 1, 36–47.
7. ———, *Knots and links*, Cambridge University Press, Cambridge, 2004.
8. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
9. Gwen Fisher and Blake Mellor, *On the topology of Celtic knot designs*, Bridges: Mathematical Connections in Art, Music, and Science (Winfield, Kansas), 2004, pp. 37–44.
10. Felix Flicker, Steven H. Simon, and S. A. Parameswaran, *Classical dimers on Penrose tilings*, Physical Review X **10** (2020), 011005.
11. Uli Gaenshirt, *Ammann grid and knot structure of a quasiperiodic girih pattern*, Proceedings of Bridges 2024: Mathematics, Art, Music, Architecture, Culture, 2024, pp. 155–162.
12. Martin Gardner, *Penrose tiles to trapdoor ciphers*, W. H. Freeman, New York, 1989.
13. Paulus Gerdes, *Lunda geometry: Mirror curves, designs, knots, polyominoes, patterns, symmetries*, second ed., Lulu, Morrisville, NC, 2007.
14. Chaim Goodman-Strauss, *Aperiodic hierarchical tilings*, Foams and Emulsions (J. F. Sadoc and N. Rivier, eds.), NATO Science Series E, vol. 354, Kluwer Academic Publishers, Dordrecht, 1999, pp. 481–496.
15. Jonathan L. Gross and Thomas W. Tucker, *A Celtic framework for knots and links*, Discrete & Computational Geometry **46** (2011), 86–99.
16. Branko Grünbaum and G. C. Shephard, *Tilings and patterns*, second ed., Dover, 2016.
17. W. Patrick Hooper, *Truchet tilings and renormalization*, arXiv:1103.6046, 2011.
18. ———, *Renormalization of polygon exchange maps arising from corner percolation*, Inventiones Mathematicae **191** (2013), no. 2, 255–320, arXiv:1105.6137.
19. Slavik Jablan, Ljiljana Radović, Radmila Sazdanović, and Ana Zeković, *Knots in art*, Symmetry **4** (2012), no. 2, 302–328.
20. Lean FRO, *Comparator: a trustworthy judge for Lean proofs*, `https://github.com/leanprover/comparator`, tag `v4.34.0` (accessed October 2026).
21. ———, `leanchecker`, part of the Lean 4 toolchain from version 4.28.0, `https://github.com/leanprover/lean4`; formerly `https://github.com/leanprover/lean4checker`.
22. Lucy Liu, Gary P. T. Choi, and L. Mahadevan, *Quasicrystal kirigami*, Physical Review Research **4** (2022), no. 3, 033114, arXiv:2104.13399.
23. Samuel J. Lomonaco and Louis H. Kauffman, *Quantum knots and mosaics*, Quantum Information Processing **7** (2008), 85–115, arXiv:0805.0339.
24. Stefan Pautze, *Space-filling, self-similar curves of regular pentagons, heptagons and other n-gons*, Proceedings of Bridges 2021: Mathematics, Art, Music, Architecture, Culture, 2021, pp. 157–164.
25. Gábor Pete, *Corner percolation on $\mathbb{Z}^2$ and the square root of 17*, Annals of Probability **36** (2008), no. 5, 1711–1747.
26. Shobhna Singh, Jerome Lloyd, and Felix Flicker, *Hamiltonian cycles on Ammann–Beenker tilings*, Physical Review X **14** (2024), 031005.
27. David Smith, Joseph Samuel Myers, Craig S. Kaplan, and Chaim Goodman-Strauss, *A chiral aperiodic monotile*, Project page, `https://cs.uwaterloo.ca/~csk/spectre/`, 2023.
28. ———, *An aperiodic monotile*, Combinatorial Theory **4** (2024), no. 1, Paper No. 6, arXiv:2303.10798.
29. ———, *A chiral aperiodic monotile*, Combinatorial Theory **4** (2024), no. 2, Paper No. 13, arXiv:2305.17743. Theorem, figure, section and page numbers refer to the published version.
30. Joshua E. S. Socolar and Joan M. Taylor, *An aperiodic hexagonal tile*, Journal of Combinatorial Theory, Series A **118** (2011), no. 8, 2207–2231, arXiv:1003.4279.
31. Simon Tatham, *Finite-state transducers for substitution tilings*, arXiv:2512.16595v2, 2025.

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